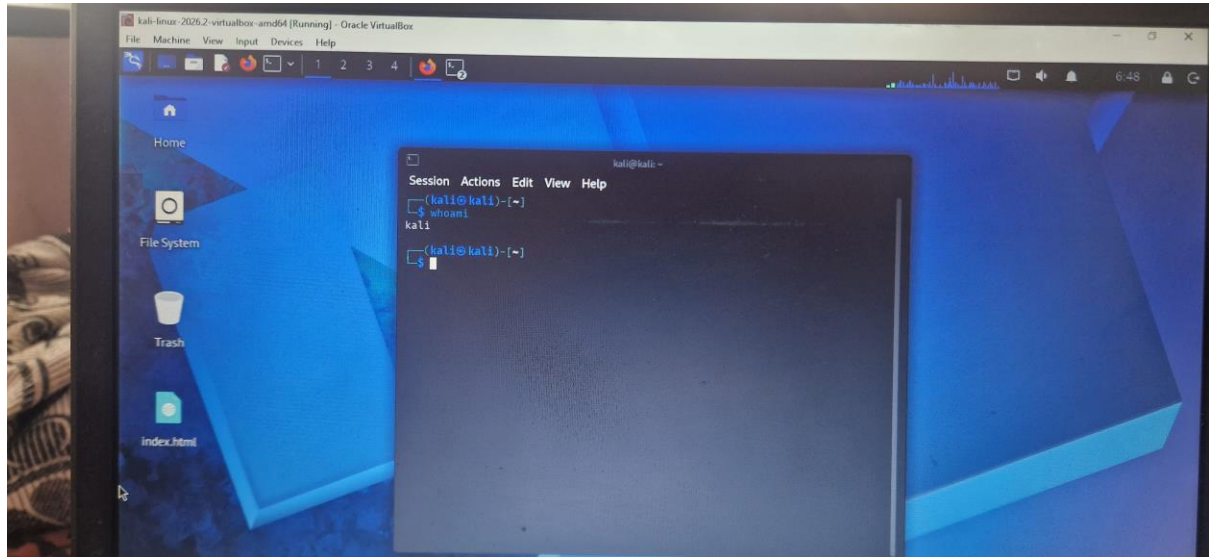
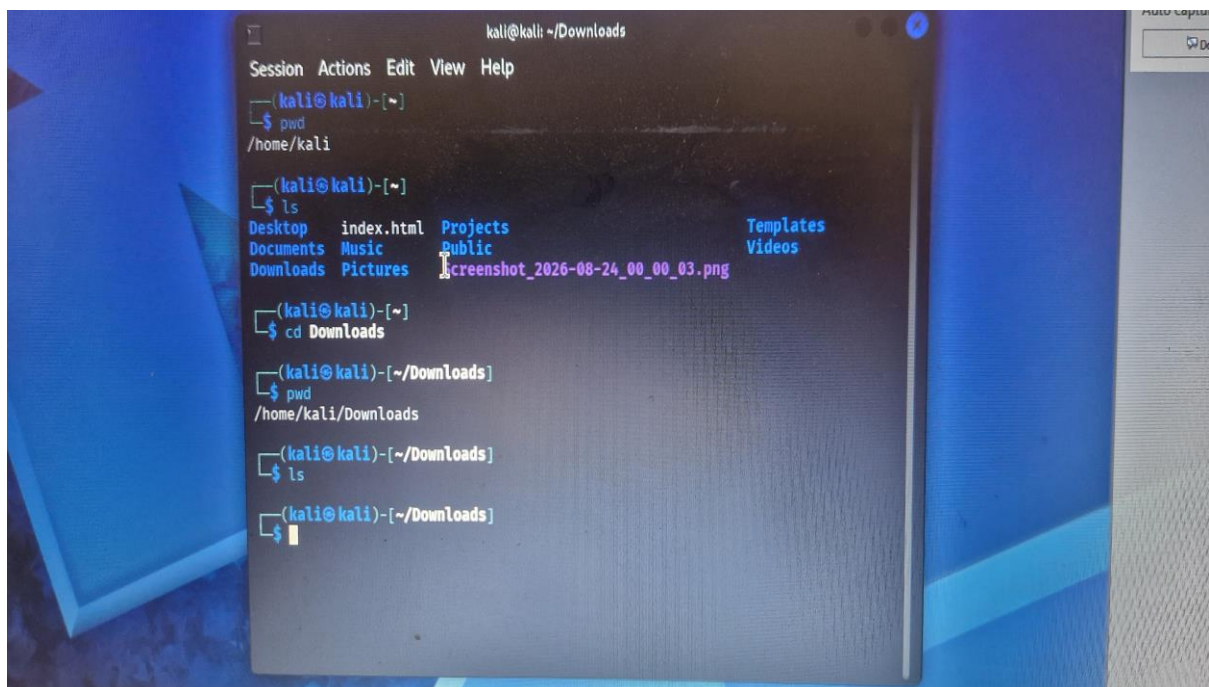


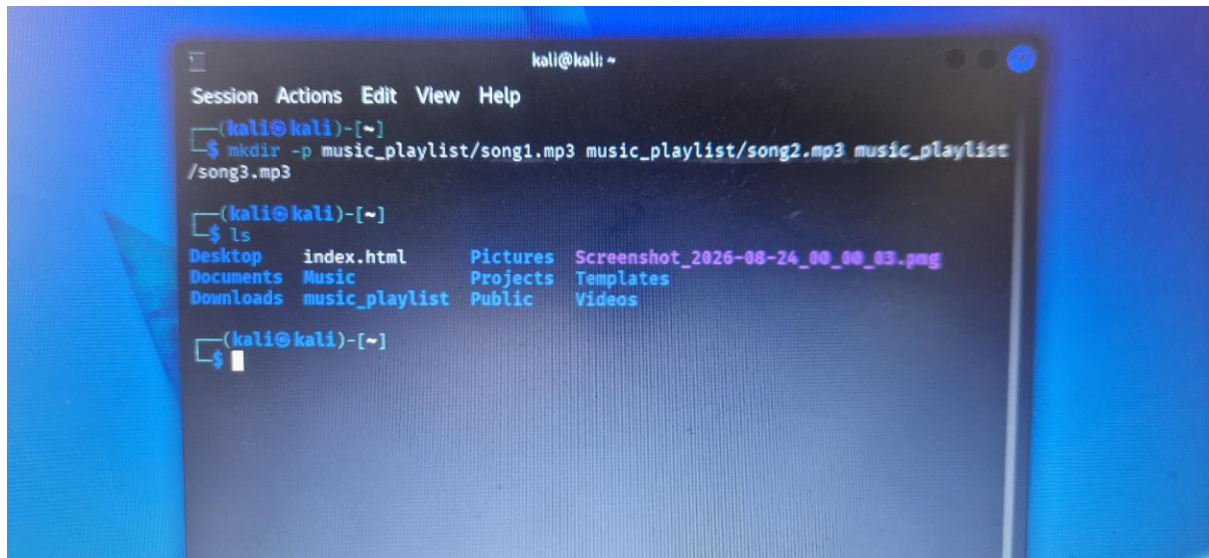
1. Install Ubuntu (or any Linux distribution) on your system using a virtual machine tool like VirtualBox, and take a screenshot of your successful login to the Linux terminal.



2. Open the Linux terminal and use the commands pwd, ls, and cd to navigate to your Downloads folder, then list all files present in it.



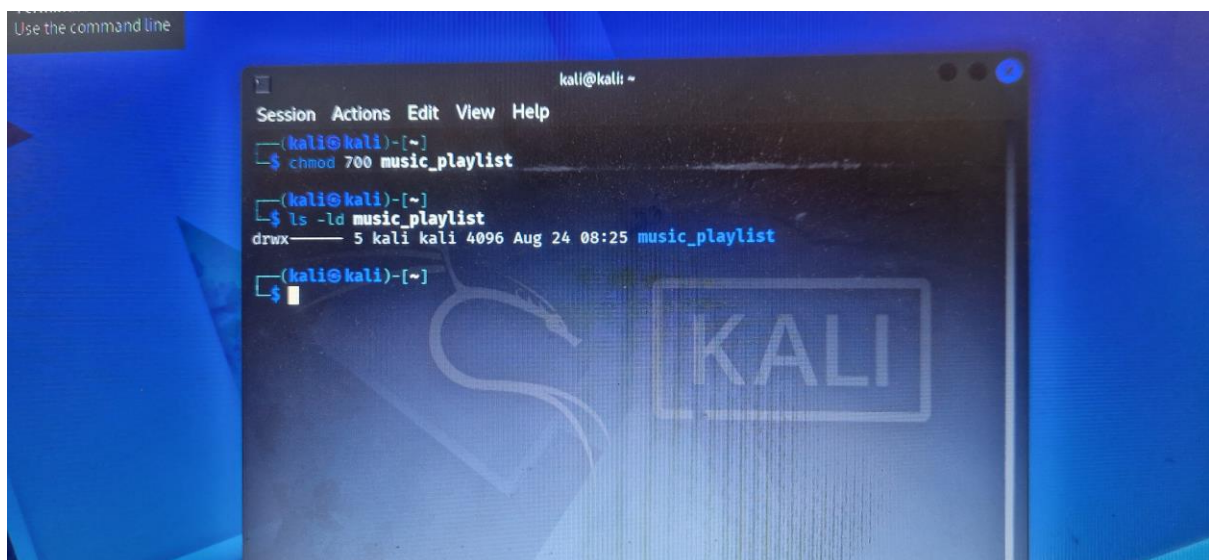
3. Create a new directory named 'music playlist' inside your home directory, and inside it, create three empty files named song1.mp3, song2.mp3, and song3.mp3 using a single Linux command.



A terminal window on a Kali Linux desktop. The prompt is (kali@kali)-[~]. The command `mkdir -p music_playlist/song1.mp3 music_playlist/song2.mp3 music_playlist/song3.mp3` is entered and executed. Below the command, the output of `ls` is shown, listing the contents of the home directory: Desktop, Documents, Downloads, index.html, Music, music_playlist, Pictures, Projects, Public, Screenshot_2026-08-24_00_00_03.png, Templates, and Videos.

```
(kali@kali)-[~]
$ mkdir -p music_playlist/song1.mp3 music_playlist/song2.mp3 music_playlist/song3.mp3
(kali@kali)-[~]
$ ls
Desktop    index.html  Pictures    Screenshot_2026-08-24_00_00_03.png
Documents  Music       Projects    Templates
Downloads  music_playlist Public       Videos
```

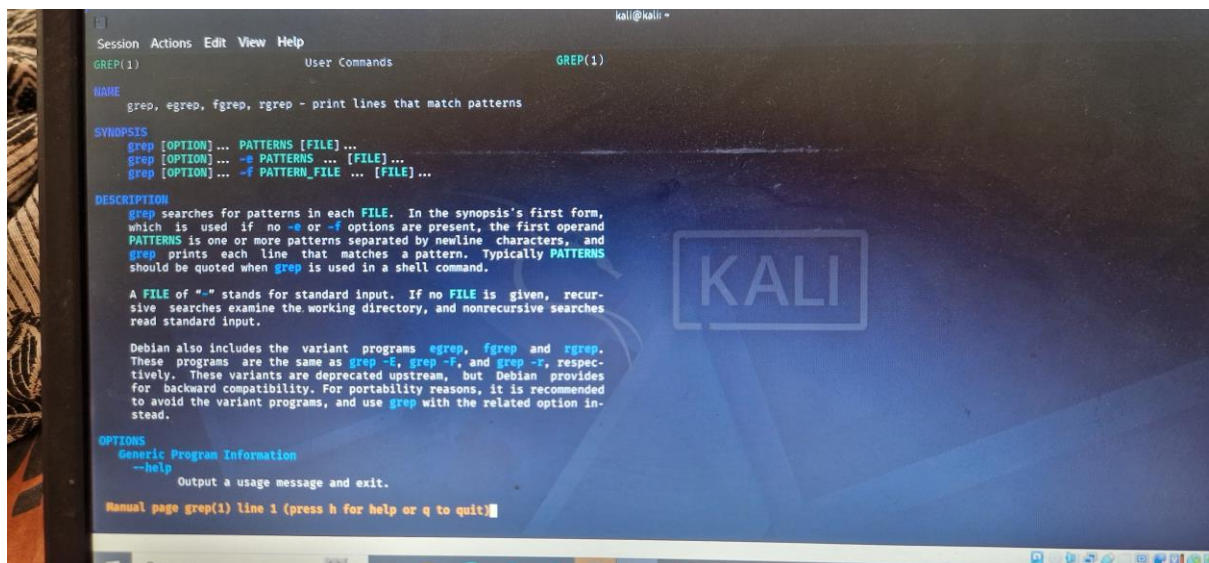
4. Change the permissions of the 'music playlist' directory so that only you (the owner) have read, write, and execute permissions, and no one else has any permissions.



A terminal window on a Kali Linux desktop. The prompt is (kali@kali)-[~]. The command `chmod 700 music_playlist` is entered and executed. Below the command, the output of `ls -ld music_playlist` is shown, displaying the permissions `drwx` for the directory.

```
(kali@kali)-[~]
$ chmod 700 music_playlist
(kali@kali)-[~]
$ ls -ld music_playlist
drwx----- 5 kali kali 4096 Aug 24 08:25 music_playlist
(kali@kali)-[~]
$
```

5. Use the man command in the Linux terminal to find out what the 'grep' command does, and write a one-line summary in your own words.

A screenshot of a Kali Linux terminal window displaying the man page for the 'grep' command. The terminal has a dark background with a 'KALI' logo. The man page text is as follows:

```
Session Actions Edit View Help
GREP(1)                                User Commands                                GREP(1)

NAME
grep, egrep, fgrep, rgrep - print lines that match patterns

SYNOPSIS
grep [OPTION]... PATTERNS [FILE]...
grep [OPTION]... -e PATTERNS ... [FILE]...
grep [OPTION]... -f PATTERN_FILE ... [FILE]...

DESCRIPTION
grep searches for patterns in each FILE. In the synopsis's first form,
which is used if no -e or -f options are present, the first operand
PATTERNS is one or more patterns separated by newline characters, and
grep prints each line that matches a pattern. Typically PATTERNS
should be quoted when grep is used in a shell command.

A FILE of "-" stands for standard input. If no FILE is given, recur-
sive searches examine the working directory, and nonrecursive searches
read standard input.

Debian also includes the variant programs egrep, fgrep and rgrep.
These programs are the same as grep -E, grep -F, and grep -R, respec-
tively. These variants are deprecated upstream, but Debian provides
for backward compatibility. For portability reasons, it is recommended
to avoid the variant programs, and use grep with the related option in-
stead.

OPTIONS
Generic Program Information
--help
    Output a usage message and exit.

Manual page grep(1) line 1 (press h for help or q to quit)
```

Grep searches text for lines that match a specified pattern or word and displays the matching lines.